

DocInsight AI Report

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AI Summary

This systematic review, authored by Manju Sharma, Vipin Kumar, and Sandeep Kumar, examines the complex relationship between rapid urban sprawl and land use/land cover (LULC) changes in India. As one of the world's fastest-urbanizing nations, India faces significant environmental challenges, with projections suggesting it will host a vast percentage of the global urban population by 2050. The study employs a multi-stage sampling methodology to analyze 58 peer-reviewed journal articles published between 1981 and 2022. The research identifies rapid, unplanned urban expansion as the primary driver behind LULC changes, which contribute to irreparable environmental degradation, including disruptions in natural geochemical cycles, biodiversity loss, and increased vulnerability to natural disasters. The authors emphasize that current urbanization patterns often prioritize immediate human desires over long-term ecological stability. A key finding is that 77.58% of the reviewed studies highlight the effectiveness of geospatial technology, numerical modeling, and remote sensing tools in accurately monitoring and measuring sprawl. Consequently, the study concludes that urban planners and government administrators must adopt these advanced scientific techniques to manage growth in its early stages. The review ultimately advocates for a shift toward sustainable urbanization policies that integrate rigorous data analysis to mitigate the negative impacts of accidental urban growth on the natural landscape and ensure environmental health.

Keywords

urban, remote sensing, urban sprawl, urban growth, land, land cover, Landsat, City, Indian Society, India